Guidelines and external resources

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Examples of good dissertations

See the archive of outstanding project dissertations (mark 80% and above).

Project resources

Learning resources

There are a number of useful resources related to honours projects available from the IAD. See the dissertation page (University webpage) including a useful project and dissertation planner. There is also advice available on:

- Statistics
  (alternatively, see this short introductory course to statistics for advanced Informatics students)
- Presentations
- Editing and proofreading

Computing resources

- Cluster computing resources (e.g. the teaching cluster (mlp) for UG/MSc students)

General advice on reading and writing

These are external resources.

- How to read a paper (alternative link)
- What is a literature review?
- How to write a literature review
- Advice on writing UG4 and MSc project reports (see background section)
- Short guides on referencing and plagiarism
- Linguistics, Style and Writing in the 21st Century - Steven Pinker
  An entertaining lecture on a simple conceptual framework and practical tips on improving your writing.
- Bailey, S. (2015). Academic writing a handbook for international students (Fourth ed.). London: Routledge

Guidance on report writing

These comments do not correspond directly to categories of the marking form - they are mainly based on previous experience in marking projects.

Books on writing

Here are some recommendations from Philip Wadler on writing.

- *The Elements of Style*, William Strunk and E.B. White.
  [Main Library (STANDARD LOAN) - 3rd floor PE1421 Str.]
- *The Sense of Style: the thinking person's guide to writing in the 21st century*, Steven Pinker.
  [Main Library (HUB SHORT LOAN) - Ground floor PE1421 Pin.]
- *The Complete Plain Words*, Sir Ernest Gowers.
  [Main Library (STANDARD LOAN) - 3rd floor PE 1421 Gow 1986]
- *On Writing Well: An informal guide to writing non-fiction*, William Zinsser.
  [Main Library (STANDARD LOAN) - 3rd floor PE1429 Zin.]
- *Mathematical Writing*: Donald E. Knuth, Tracy L. Larrabee, and Paul M. Roberts
  Washington, D.C.: Mathematical Association of America, 1989, ii+115pp. PDF(external link)
  [Minicourse on Technical Writing, pp.1—6, is especially recommended.]

Many editions of these books, both new and used, are available for sale in bookshops and online.

Talks on technical writing

- Murdoch J. Gabbay, "On technical writing", 2020. (pdf)

One-to-one practical sessions on academic writing by Royal Literary Fund Fellows

Visit this link (external link).

Advice from supervisors (part 1)

Here are a few comments by Mary Cryan, former UG4 project supervisor, with some additions by Helen Pain, Don Sannella, and Stephen Gilmore.

- The project is marked on your report, backed up by your code. If you have done a superb job of system-building, and submit a bad report, you risk getting a low score. So please make sure you put sufficient effort into delivering a good report. Ensure that you stop work on your implementation early enough to be able to write up your report well.

- If you absolutely cannot decide how to structure your report, then having the following five chapters will usually work out okay: Introduction, Background, Design, Implementation, Evaluation and Conclusions. Your report does not have to have this structure, not at all, but it is a reasonable place to start for most projects.
  - Consider also having a chapter on Testing before Evaluation; the difference being that testing is done by you as the system developer, whereas evaluation is done by others in user testing.
- The Introduction chapter should always provide a 'roadmap' to the report. Of course it should provide an introduction to the problem being considered, but it should also give some details of what you did - do not leave this to the conclusion. You should give forward references into the rest of the report - e.g.,
  - "In Chapter 2 how algorithms and heuristics are used to deal with approximate counting are discussed",
  - "The design of the system is presented in Chapter 4",
  - "In Chapter 3 the reasons for choosing to focus on the bounded-degree case of this problem are explained".

- A common problem in many dissertations is having too much in the Background chapter. Don't put in everything that you learned in order to do your project; instead put in only the things that the reader needs to know in order to appreciate your dissertation.

- **Structure**: Please use the environments available in LaTeX to structure your report. Use `subsection` and `subsubsection` to separate out the topics of each chapter. Important points should also be displayed so that they are more visible than ordinary text (e.g., using the `itemize` or the `quote` environment). The reader should not have to read deep into a page of text to pick out a key fact.

- Through the report, spell out what was **done** (past tense). Make clear what was your contribution to the task. If you are building on an existing system, or using libraries that already exist, make clear the division between your work and existing work.

- If your project is being done in tandem with another student, be very clear what is your work and what is their work. It is helpful if there is an easy-to-explain separation between your work and the work of the other student (e.g. front-end/back-end, or software/hardware). Be sure to tell the reader what this separation is.

- Your report will be read by markers throughout the School of Informatics. Although you can generally assume that your second-marker is knowledgeable about informatics, do not assume that they are an expert in your subfield. Give some basics as well as the details. Also, make sure you point out what was involved in solving certain problems; this can help a non-expert judge the work involved. For instance:
  - "The interpolation algorithm was implemented in C directly, rather than using the routines in Matlab".
  - "In order to develop a user interface appropriate for the educational system, four prototypes were tested on 50 students".
- **Tenses**:
  - Many of you use your interim report as part of your final report. If you do this, please change the tenses - from "will .. " to "did ...". It is important to be careful with this as not all tasks proposed before Christmas always get completed.
  - Also, try not to use the "generic tense", e.g. "CoolApp Systems use hatterbash technologies for preprocessing of rino-data", **if you are talking about your own project**. Be clear what was done in the particular case of your work. Instead write, "As is common in CoolApp systems, hatterbash technologies were used to preprocess the rino-data" (and continue giving details ...). Generic tense is fine for the background chapter.

- **Voice**: It is often recommended that reports (and research papers in general) are written in the passive voice, as opposed to active voice, e.g., the above, rather than: "As is common in CoolApp systems, I used hatterbash technologies to preprocess my rino-data". The reason for this is that less experienced writers can sometimes find it be harder to be objective when writing in the active voice. However, it is up to you (and the recommendations of your supervisor) whether you prefer to write in the active or passive voice. Some readers find the use of passive voice to be dull and unengaging.

- Prefer clarity to formality. If you find yourself using words like whilst or henceforth in a sentence you should perhaps hit the delete key on that one and try to express yourself more simply and directly.

- Avoid run-on sentences. As a general rule of thumb, your sentences in your early drafts of your dissertation will likely be too long. When editing, try to split over-long sentences into shorter, more digestable ones.
- Avoid contractions. Instead of isn't, don't, we're, write is not, do not, we are. Quotations are an exception to this rule. For example, The Rolling Stones sang, "You can't always get what you want."
- Your source code. It is helpful for the marker to be able to look at your source, or other type of documentation, as a back-up to the report. You are required to maintain a subdirectory of your DICE account where you store source code, or processed data, or any supporting evidence which is appropriate to your project. You will be required to submit this entire directory along with your report.
- If there was some technical challenge that you could not resolve during the period of the project, it will still be useful to devote space in your report to discussing a design (for example) that did not get implemented, or an implementation that was not completed.
- It is okay to get your classmates to read over your work for grammar, punctuation, feedback on clarity, etc. But they should not contribute in any way to the technical content of the report.

Advice from supervisors (part 2)

Advice from Philip Wadler:

- Think about the best order in which to present information. Don't just write it all down in the order it occurs to you.
- At each point, keep your reader in mind. What do they know already? What will they want to learn next?
- Don't just say: first I did this, then I did that. At each choice point, explain the alternatives and why you chose the alternative you did. Bonus points if you tie the reason to principles you've learned.
- Chronological may not be the best order: iterating through a series of solutions can be repetitive and tedious. The reader may not care about the order in which you did your work. Instead, present your final solution, focussing on key points.
- Don't present code, documentation, theorems, or formulas as a blob of material for the reader to decypher. Introduce each blob with a single-sentence description of its purpose. For instance:

  BAD. We will need the following.

max x y = if x ≥ y then x else y

GOOD. We will need to compute the maximum of two numbers.

max x y = if x ≥ y then x else y

- In a good story, there is a surprise at the end. In a good report, it is the opposite: all important results should be up front. After the marker has read your introduction, they should know what you've done and what mark they will give you (assuming the report backs up the claims in the introduction).
- "Particulars, as every one knows, make for virtue and happiness; generalities are intellectually necessary evils."
  — Aldous Huxley, Brave New World.

Awards

Skyscanner Accessibility and Inclusion Award

The ***Skyscanner Accessibility and Inclusion Award*** aims to acknowledge and honour student efforts in promoting disability inclusion through their dissertations. Supervisors are invited to nominate students who have addressed accessibility and/or inclusion challenges in their UG/MInf dissertations. Alternatively, students can also nominate themselves. For details, see Skyscanner Accessibility and Inclusion Award (external link).